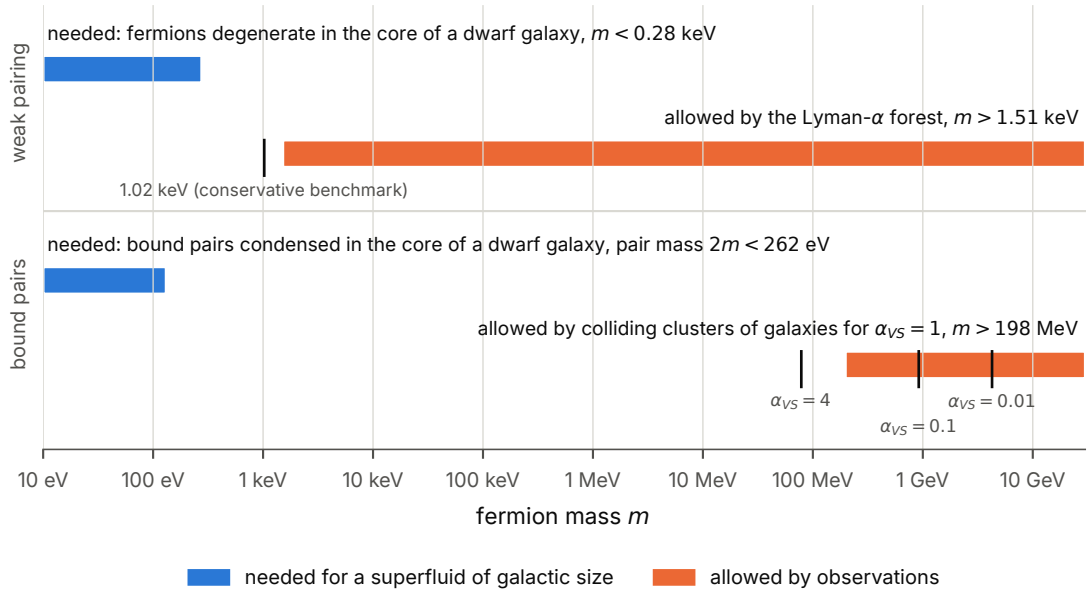
weak pairing
needed: fermions degenerate in the core of a dwarf galaxy, $m < 0.28$ keV
allowed by the Lyman-α forest, $m > 1.51$ keV
1.02 keV (conservative benchmark)
bound pairs
needed: bound pairs condensed in the core of a dwarf galaxy, pair mass $2m < 262$ eV
allowed by colliding clusters of galaxies for $\alpha_{VS} = 1$, $m > 198$ MeV
$\alpha_{VS} = 4$
$\alpha_{VS} = 0.1$
$\alpha_{VS} = 0.01$
10 eV
100 eV
1 keV
10 keV
100 keV
1 MeV
10 MeV
100 MeV
1 GeV
10 GeV
fermion mass m
needed for a superfluid of galactic size
allowed by observations